

# Tensors and their Geometry

Vincenzo Antonio Isoldi

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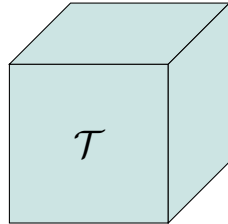


**CBG**  
Max Planck Institute  
of Molecular Cell Biology  
and Genetics

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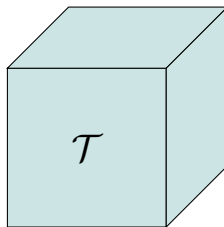
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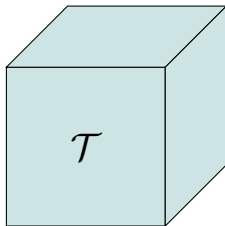
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- ▶ An element of a tensor product of vector spaces:  $\mathbb{C}^{n_1} \otimes \dots \otimes \mathbb{C}^{n_d}$ .
- ▶ A multilinear map (once bases are fixed).
- ▶ **A point of a suitable space.**

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Note that  $\mathbb{C}^{n_1} \otimes \dots \otimes \mathbb{C}^{n_d} \simeq \mathbb{C}^{n_1 \times \dots \times n_d}$ .

# Tensor Rank and Tensor Rank Decomposition

Consider a tensor  $\mathcal{T} \in \mathbb{C}^{n_1} \otimes \cdots \otimes \mathbb{C}^{n_d}$ . Among them, there are some particularly easy ones, called **rank-1** tensors, i.e., tensors that can be written as

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The **tensor rank decomposition** (TRD) writes a tensor

$$\mathcal{A} \in \mathbb{C}^{n_1} \otimes \cdots \otimes \mathbb{C}^{n_d}$$

as a minimum-length linear combination of simple tensors:

$$\mathcal{A} = \sum_{i=1}^r \alpha_i \mathbf{a}_i^1 \otimes \cdots \otimes \mathbf{a}_i^d, \quad \mathbf{a}_i^k \in \mathbb{C}^{n_k}, \quad \alpha_i \in \mathbb{C}.$$

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## Definition

The least integer for which  $\mathcal{A}$  admits such a decomposition is called the **rank** of  $\mathcal{A}$ . It is denoted by  $\text{rank}(\mathcal{A})$ .

# Motivating Example: The Cocktail Party Problem



Suppose we have

- $r$  people in the room, speaking, and
- $r$  microphones spread over the room at fixed locations.

Can we figure out who is saying what?

# Motivating Example: The Cocktail Party Problem

Because of the superposition of sound waves, we write

$$Y = MX,$$

where

- row  $i$  of  $Y$  contains the sound measured at microphone  $i$ ;
- row  $i$  of  $X$  contains the sound produced by person  $i$ ; and
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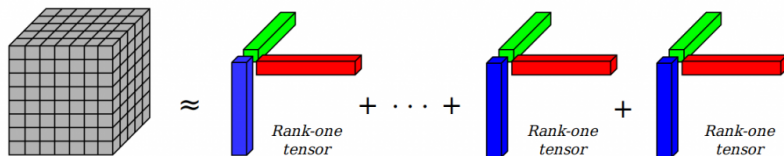
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Can we recover the individual speakers in  $X$  by recording  $Y$  ? Of course not! This problem is not identifiable because

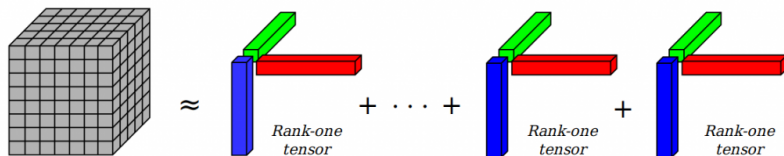
$$Y = MX = (MA^{-1})(AX) = M'X'$$

for all invertible  $A$ .

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| Algorithm                       | Type          | Noise Robustness | Speed     |
|---------------------------------|---------------|------------------|-----------|
| ALS (Alternating Least Squares) | Iterative     | Medium           | Variable  |
| Levenberg-Marquardt             | Optimization  | High             | Slow      |
| GEVD (Algebraic)                | Algebraic     | Low              | Fast      |
| Bayesian Methods                | Probabilistic | Very High        | Very Slow |

**MANY limitations:** most of these techniques require the rank of the tensor as input, but finding the rank is an **NP-hard** problem.

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Instead of looking at the actual entries of a tensor, let's look at their **structure**. We consider tensors up to scaling, identifying them as points in a projective space:

$$\mathcal{T} \in \mathbb{P}(\mathbb{C}^{n_1} \otimes \dots \otimes \mathbb{C}^{n_d}) \simeq \mathbb{P}_{\mathbb{C}}^N$$

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## The Fundamental Question

What is the geometric shape formed by the “simplest” tensors, if any?

# The Segre Variety (General Tensors)

Consider the map:

$$\begin{aligned} \text{Seg}_d : \mathbb{P}(V_1) \times \dots \times \mathbb{P}(V_d) &\longrightarrow \mathbb{P}(V_1 \otimes \dots \otimes V_d) \\ ([v_1], \dots, [v_d]) &\longmapsto [v_1 \otimes \dots \otimes v_d] \end{aligned}$$

$\mathcal{S}_d := \text{Im}(\text{Seg}_d) =$  rank-1 tensors of order  $n$ .

## Theorem

The set  $\mathcal{S}_d$  is indeed an algebraic variety, called **Segre Variety**.

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In principle, to show that  $\mathcal{S}_d$  is an algebraic variety, we have to find a (homogeneous) ideal  $I \in \mathbb{C}[x_0, \dots, x_k]$  such that  $V(I) = \mathcal{S}_d$ . In other words, we have to find (a finite number of) equations that vanish exactly on  $\mathcal{S}_d$ . Finding these equations is rather simple.

$$I(\mathcal{S}_d) = ((2 \times 2) - \text{minors of the flattenings of a generic tensor of order } d).$$

# Example

$$\begin{aligned} Seg_2 : \mathbb{P}(\mathbb{C}^2) \times \mathbb{P}(\mathbb{C}^2) &\longrightarrow \mathbb{P}(\mathbb{C}^2 \otimes \mathbb{C}^2) \simeq \mathbb{P}(\mathbb{C}^4) = \mathbb{P}_{\mathbb{C}}^3 \\ ([x_0, x_1], [y_0, y_1]) &\longmapsto [(x_0, x_1) \otimes (y_0, y_1)] \end{aligned}$$

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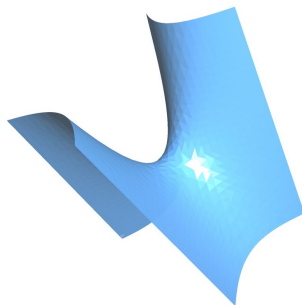
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# From Rank 1 to Rank $r$

If rank 1 tensors are points on the variety  $\mathcal{S}_d$ , what are rank  $r$  tensors?

$$\text{rank}(\mathcal{T}) = r \iff \mathcal{T} = \sum_{i=1}^r P_i, \quad P_i \in \mathcal{S}_d$$

Geometrically, the sum of points corresponds to the linear span.

- **Rank 2:** Points on **secant lines** connecting two points of  $\mathcal{S}_d$ .
- **Rank 3:** Points on **secant planes** spanning three points of  $\mathcal{S}_d$ .

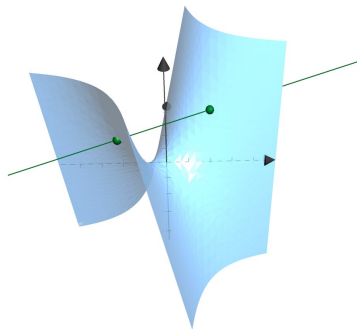
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# Secant Varieties and Border Rank

We define the  $r$ -th **Secant Variety**:

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Unlike matrices, the set of  $\text{rank} \leq r$  tensors is not always closed!

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- ▶ **Secant points:** Tensors of rank  $\leq r$ .
- ▶ **Limit points:** Tensors of **border rank**  $\leq r$ .

Geometrically, limit points lie on **tangent lines** (or tangent spaces) to the variety  $\mathcal{S}_d$ . A tensor can have rank 3 but lie on the tangent line of the Segre variety (border rank 2).

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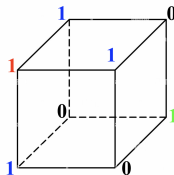
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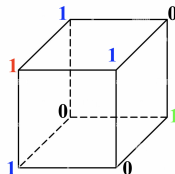
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**Example:** A rank-1 symmetric tensor corresponds to a power of a linear form:

$$v \otimes v \otimes v \longleftrightarrow L(x)^3 = (a_0x_0 + \dots + a_nx_n)^3$$

# The Veronese Variety (Symmetric Tensors)

Just as we had the Segre map for general tensors, we may consider

$$\begin{aligned}\nu_d : \mathbb{P}(V) &\longrightarrow \mathbb{P}(\mathrm{Sym}^d(V)) \\ [v] &\longmapsto [v^{\otimes d}] \quad (\text{or } [L^d])\end{aligned}$$

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Asking for  $\mathcal{T}$  to be symmetric means working on the hyperplane  $H : z_2 = z_1$ . Hence, rank-1 symmetric tensors of  $\mathbb{C}^2 \otimes \mathbb{C}^2$  can be seen as points of the quadric  $X \in \mathbb{P}_{\mathbb{C}}^3$  identified by the polynomial

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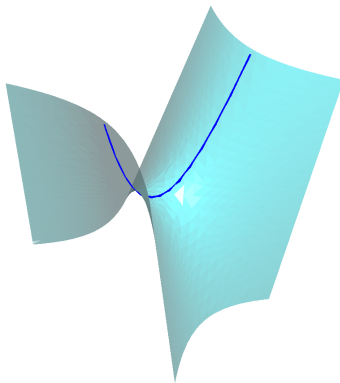
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These are points of the conic:

$$Y = \begin{cases} z_0 z_3 - z_1^2 = 0 \\ z_1 = z_2 \end{cases}.$$

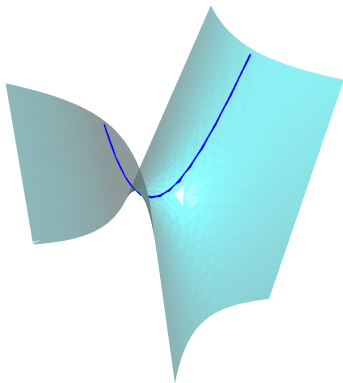
# Segre vs. Veronese

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Indeed, every rank-1 symmetric tensor is obviously a rank-1 tensor.

- **Segre:**  $v_1 \otimes v_2 \otimes \cdots \otimes v_d$  (Vectors can be different).
- **Veronese:**  $v \otimes v \otimes \cdots \otimes v$  (Vectors must be the same).

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We can recast the same question for symmetric tensors (polynomials).

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What is the minimum  $r$  such that a polynomial  $F$  can be written as:

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## Comon's Conjecture

The symmetric rank of a tensor is equal to its standard tensor rank.

## A counterexample to Comon's conjecture

Yaroslav Shitov

We present an example of a symmetric tensor of size  $800 \times 800 \times 800$  which can be written a sum of 903 simple tensors with complex entries but not as a sum of 903 symmetric simple tensors.

Comments: 13 pages; the same example as in ver. 1; section 4 corrected; section 2 split into two; more details added

Subjects: **Combinatorics (math.CO)**

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(or [arXiv:1705.08740v2](https://arxiv.org/abs/1705.08740v2) [math.CO] for this version)

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## Erratum: A Counterexample to Comon's Conjecture\*

Jan Draisma<sup>†</sup>

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**Abstract.** Shitov has discovered an error in his paper [*SIAM J. Appl. Algebra Geom.*, 2 (2018), pp. 428–443] in which he exhibits a counterexample to Comon's conjecture over the complex numbers. It is currently unknown whether this error can be corrected.

**Key words.** tensor decomposition, Comon's conjecture

**MSC codes.** 15A03, 15A69

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# Why does this Geometry matter?

## ① Uniqueness (Identifiability)

- ▶ The geometric structure of the Segre variety ensures that tensor decomposition is often **unique**.
- ▶ This is unlike matrices (SVD is unique only due to orthogonality constraints).
- ▶ In applications (chemometrics, signal processing), this uniqueness allows us to recover "ground truth" components.

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## ② Approximation

- ▶ Understanding  $\sigma_r(\mathcal{S})$  and  $\sigma_r(\nu_d)$  tells us when a best low-rank approximation exists.
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## ③ Exploiting current algorithms

- ▶ To run current algorithms, we often need to "guess" the rank of a given tensor.
- ▶ The rank of a generic tensor is given by the order of the first secant variety that fills up the ambient space.

$$X = \sigma_1(X) \subset \sigma_2(X) \subset \cdots \subset \sigma_{r-1}(X) \subset \sigma_r(X) = \mathbb{P}^N$$

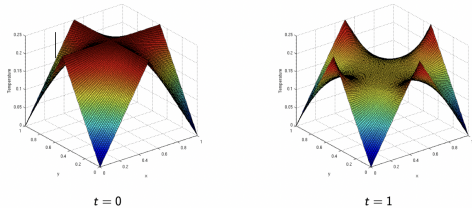
# Some Applications

Because each rank-1 summand in a tensor rank decomposition of a tensor is usually unique and always easy to interpret, the tensor rank decomposition is often used as a tool for **explorative data analysis** or a **reduced order model** of (potentially incomplete) multidimensional data.

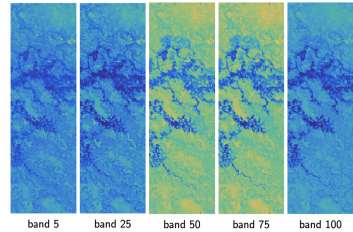
- **Factor analysis** for analyzing underlying causes in psychology (*Carroll and Chang, 1970; Harshman, 1970*)
- Identifying spectra of fluorophores in **fluorescence spectroscopy** (*Appelhof and Davidson, 1981*)
- **Estimation of the parameters of some latent variable models** like single exchangeable topic models, mixtures of Gaussians, and hidden Markov models (*Anandkumar et al., 2014*)
- Tensor-based **collaborative filtering** and **recommender systems** (*Frolov and Oseledets, 2017*)
- Reduced order model for **thermodynamic quantities of multicomponent alloys** (*Coutinho, Vervliet, De Lathauwer and Moelans, 2020*)

# Some Applications

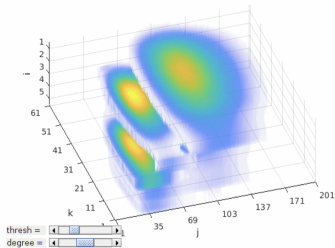
(1) Solutions of (parameterized) partial differential equations



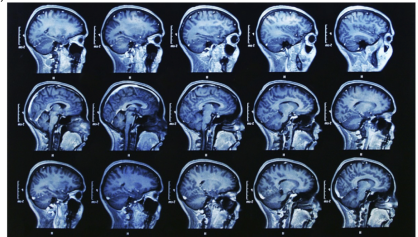
(2) hyperspectral imaging



(3) fluorescence spectroscopy



(4) MRI data



## Schur Apolarity and the Dual Terracini Lemma for Flag Varieties: A Geometric and Representation-Theoretic Approach



UNIVERSITÀ DEGLI STUDI DI TRENTO

RELATORE

Prof. Alessandra BERNARDI

CORRELATORE

Dr. Stefano CANINO

LAUREANDO

Vincenzo Antonio ISOLDI

- Talk @ CSBD - 12th March

# THANK YOU!

Let's discuss!